

Challenges in Developing Methods for Authentication of Artisan Bread and Sourdough Bread

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Nowadays sourdough bread and other sourdough-based baking goods (e.g. biscuits, crackers, pastry, pizza and pasta) are enjoying an increasing popularity as convenient, nutritious, stable, natural, low processed and healthy food. In fact, the use of sourdough in bakery industry has received a great attention worldwide in recent years, as evidenced by the advent of a diversified (but limited) number of products in market and R&D activities dedicated by companies to this topic (CA 18101 2018). It needs to be pointed out that no national legislation anywhere in the world regulates and protects traditional/typical sourdough breads. However, the use of sourdough has to be guaranteed to meet both bakery and consumer expectations, and to fulfil legal requirements (Pontonio et al. 2017).

There are a few papers published in 2000s, dealing with the topic of developing procedures for bread authentication. Brescia et al. (2003, 2007) suggest the application of nuclear-magnetic resonance and isotope-ratio mass-spectrometry methods to differentiate between Italian Protected Designation of Origin (PDO) and Protected Geographical Indication (PGI) sourdough breads obtained from tender and durum wheat flours, and to determine geographical origin of breads from southern Italy. Bianchi et al. (2008) reported a GC/MS method for the authentication of PDO Italian durum sourdough 'Altamura' bread, using volatile compounds as characteristic fingerprints.

Some studies demonstrated the potential of Alkylresorcinol levels as biomarkers for determining wholegrain wheat and/or rye content in bread using a gas chromatography method (Andersson et al. 2010), and for differentiation between breads made from whole wheat flour and refined wheat flour (Geng et al. 2016).

Other investigations report the employment of: gas chromatography – mass spectrometry technique in order to determine the content of buckwheat and wheat flour in bread (Psodorov et al. 2015), a denaturing high-performance liquid chromatography combined with single nucleotide polymorphism to trace durum wheat cultivar 'Timilia' in the bread claimed to be 'Pane Nero di Castelvetro' (Castelvetro Black Bread) (Giancaspro et al. 2016), a short-wave near-infrared hyperspectral imaging to detect adulterations in wheat flour and bread with cheap grains, such as sorghum, oats and corn (Verdu et al. 2016), and a liquid chromatography – tandem mass spectrometry as an analytical strategy to check the authenticity of wheat, spelt and rye addition to bread products, using peptide markers (Bönick et al. 2017).

Within the frame of bread authentication most published research articles report the application of a polymerase chain reaction (PCR) methods in order to verify wheat bread supplementation with maize, rye and oat flours (Yilmaz et al. 2019), to detect the presence of common soft wheat in durum wheat bread (Pasqualone et al. 2007) and the presence of demanded durum wheat cultivars in PDO 'Altamura' bread (Pasqualone et al. 2010). The research article most tightly related to the topic of authentication of sourdough breads is published by Pontonio et al.

(2017), in which the authors describe the application of a quantitative PCR method to detect the lactic acid bacterial microbiota in bread and, therefore, discriminate between breads made with and without sourdoughs.

After analyzing the research reported in this work a conclusion can be drawn that there is a serious lack in analytical methods focusing on developing modern solutions for authentication of breads, especially sourdough breads, which are lately being promoted as novel and healthier alternatives to common breads. With this in mind, there is a strong obligation for the scientific community to work on developing accurate, reliable and robust methods for verifying authenticity of sourdough bread and other bakery products, using sophisticated analytical instrumentation linked with modern statistical and data processing tools.

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